Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets "Neonics" and "Litter", respectively. Be sure to include the subcommand to read strings in as factors.

```
#Import packages
library(tidyverse)
library(lubridate)
library(here)

#Check workspace
here()
```

```
## [1] "/home/guest/Environmental Data Visualization Class/EDE_Fall2025"
```

```r
#Import ECOTOX_Neonicotinoids_Insects_raw.csv
Neonics<-read.csv(
  file=here('Data','Raw','ECOTOX_Neonicotinoids_Insects_raw.csv'),
  stringsAsFactors=T )
```

```r
#Import NEON_NIWO_Litter_massdata_2018-08_raw.csv
Litter <- read.csv(
  file=here('Data','Raw','NEON_NIWO_Litter_massdata_2018-08_raw.csv'),
  stringsAsFactors= T)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

   Answer:Researchers thought that they were a less harmful alternative to previous insecticides, but new research has shown that they have devastating ecological impacts, for example they are toxic to pollinators e.g. bees, insects that are beneficial to the environment, and aquatic invertebrates.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

   Answer: It provides habitat for a wide range of organisms, and it is important for nutrient cycling and controling hydrology and stream environments

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

   Answer: 1.Sampling is only done at sites where woodt vegetation is less than 2 cm tall 2.Sampling is only conducted in tower plots 3. As far as possible, sampling is done in the same location over the duration of the observatory.

Obtain basic summaries of your data (Neonics)

What is are the dimensions of the dataset

```r
#5.  What are the dimensions of the dataset?
  dim(Neonics)
```

```
## [1] 4623   30
```

6. Using the `summary` function on the "Effect" column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
#sort(Neonics$Effect)
sort(summary(Neonics$Effect), decreasing=TRUE)
```

```
##       Population        Mortality         Behavior Feeding behavior
##             1803             1493              360              255
##     Reproduction      Development        Avoidance         Genetics
##              197              136              102               82
##        Enzyme(s)           Growth       Morphology    Immunological
##               62               38               22               16
##     Accumulation      Intoxication     Biochemistry          Cell(s)
##               12               12               11                9
##       Physiology        Histology        Hormone(s)
##                7                5                1
```

Answer: The most common studied effects are:population, mortality, behavior, reproduction, development and avoidance

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary(Neonics$Species.Common.Name, maxsum = 7)
```

```
##            Honey Bee       Parasitic Wasp Buff Tailed Bumblebee
##                  667                  285                   183
##   Carniolan Honey Bee           Bumble Bee      Italian Honeybee
##                  152                  140                   113
##              (Other)
##                 3083
```

```
#I did maxsum= 7, to get away from having other as the 6th species
```

Answer: The six most commonly studied species are: Honey Bee, Parasitic Wasp, Buff Tailed Bumblebee, Carnolian Honey Bee, Bumble Bee, Italian Honeybee. These are all types of bees, and bees feed directly on pollen and nectar, making them especially suceptible to Neonicotinoids because these insectidicdes are sytematic and accumulate in pollen and nectar, vs other insects that might feed on other parts of the plant and have less contact with pollen and nectar, thus making them less impacted.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

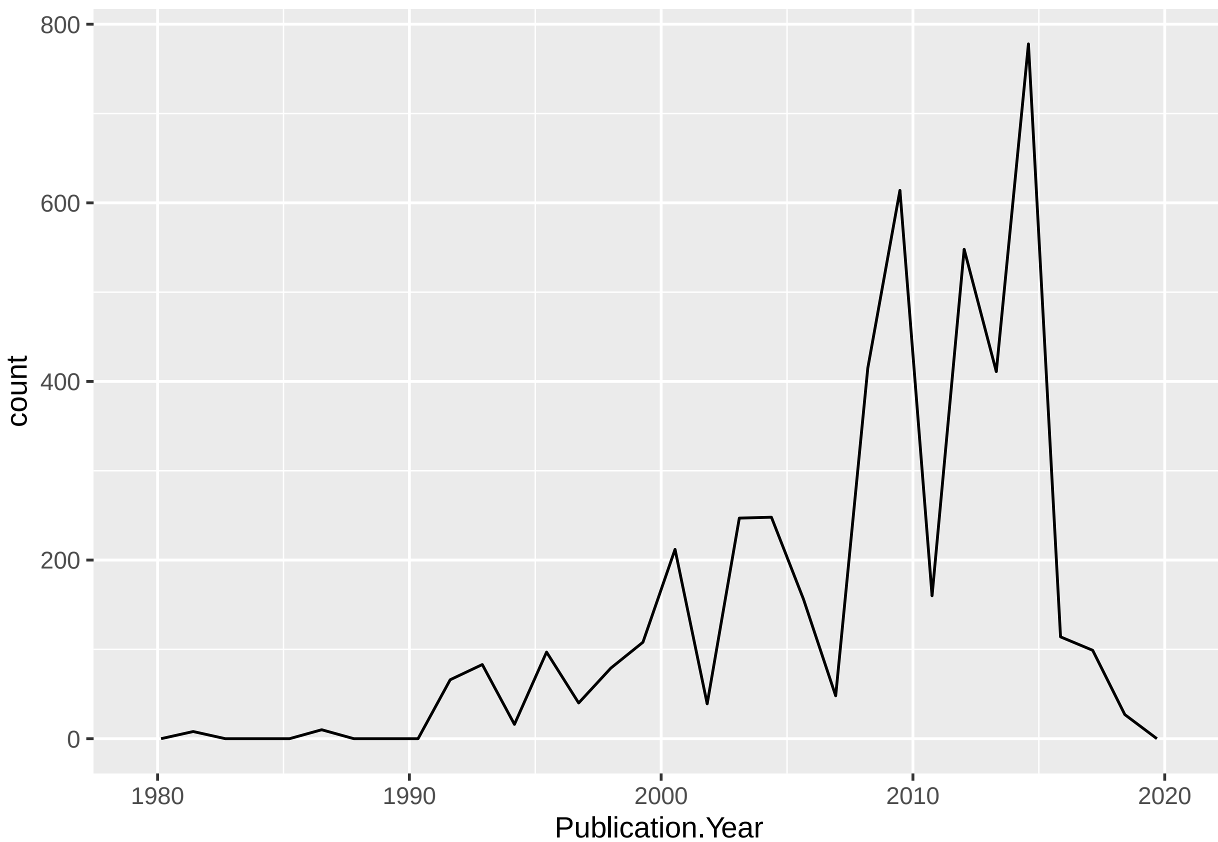
Answer:It is of the class factor because it contains numbers and characters, thats why it is not numeric (i.e. NR)

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
ggplot(Neonics) +
  geom_freqpoly(aes(x=Publication.Year))
```

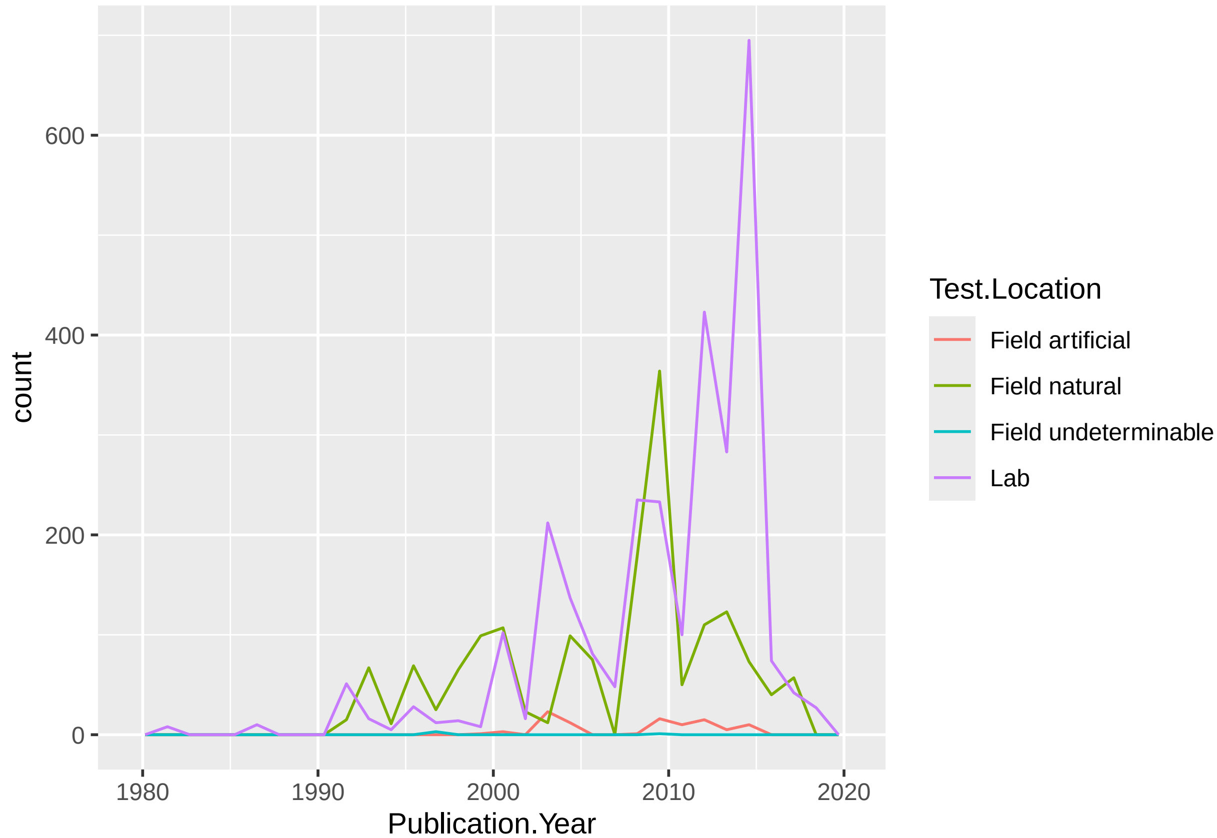
```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(Neonics) +
  geom_freqpoly(aes(x=Publication.Year, color=Test.Location))
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



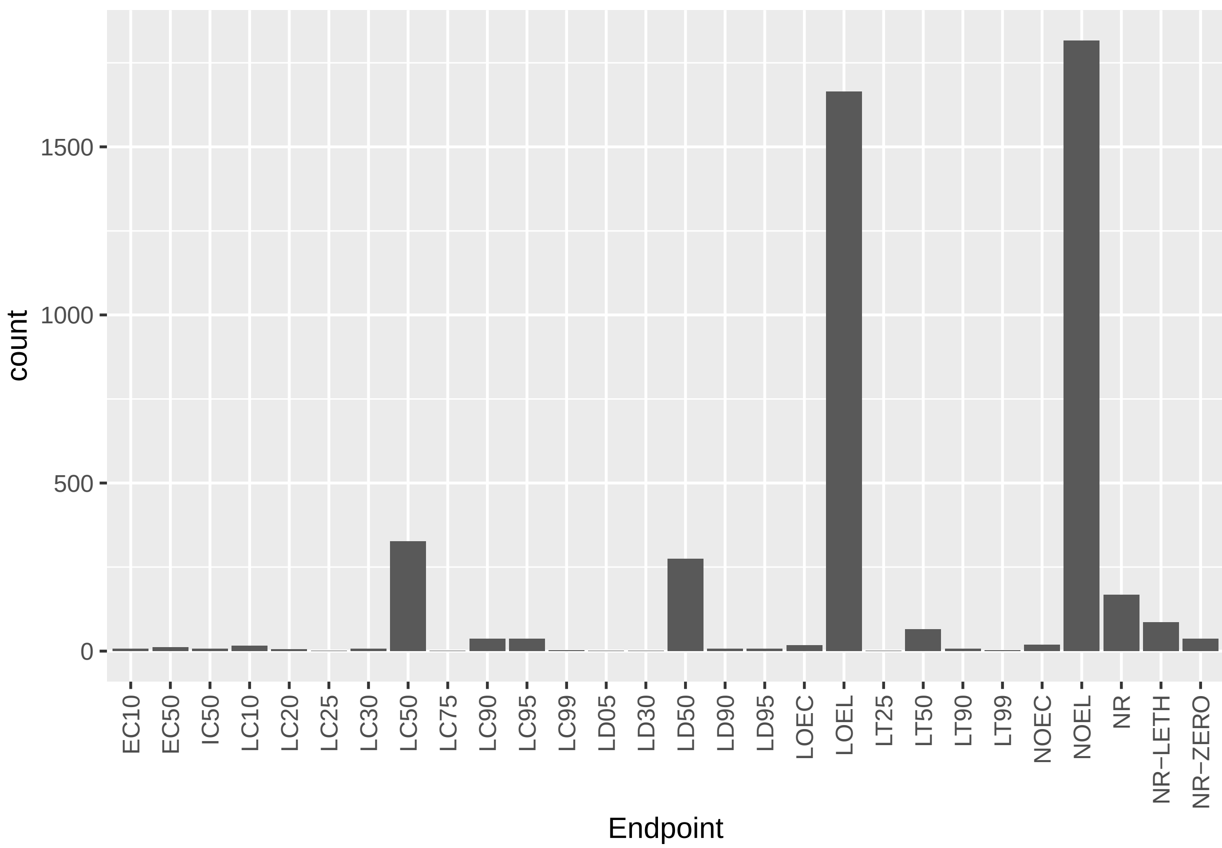
Interpret this graph. What are the most common test locations, and do they differ over time?

> Answer:Field artifical and field natural are the most common test locations, and the general trend is that they seem to increase sharply over time between 2005 and 2015, but then it has been decreasing since about 2018.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP**: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels…]

```
ggplot(data=Neonics, aes(x=Endpoint)) +
  geom_bar() + theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The two most common points are LOEL- lowest observed effect level, meaning this is the lowest dose or exposure level of a substance at which an observable effect is detected in a test population; and NOEL-The highest dose or exposure level of a substance that produces no observable adverse effects in a test population.

Explore your data (Litter)

12. Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
#Checking class of collectDate
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
#Since it was not date, I changed it and confirmed that it wwas changed

date_of_collection <- ymd(Litter$collectDate)
class(date_of_collection)
```

```
## [1] "Date"
```

```r
unique(date_of_collection)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```r
unique(Litter$plotID)
```

```
##  [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
##  [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

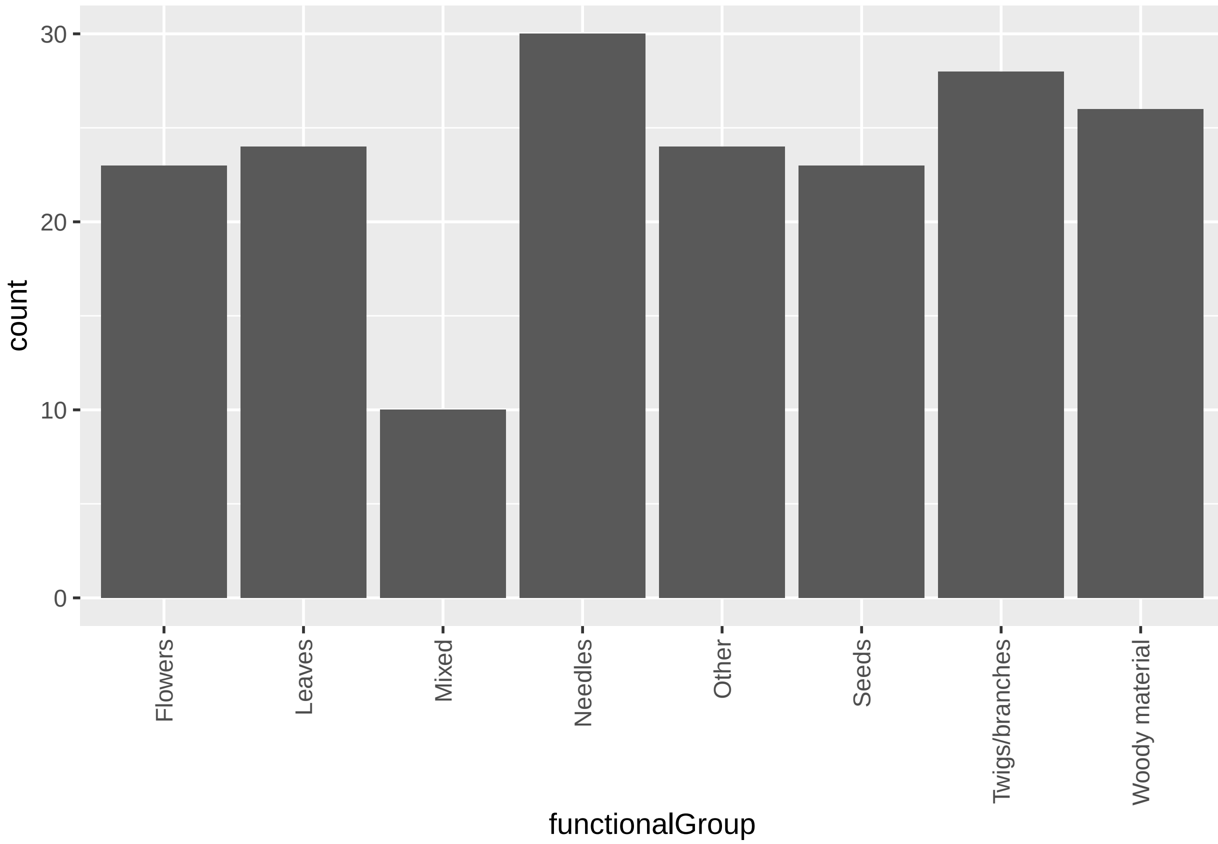
```r
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061
##       20       19       18       15       14        8       16       17
## NIWO_062 NIWO_063 NIWO_064 NIWO_067
##       14       14       16       17
```

> Answer:Unique gives you a list of distinct plot IDS where sampling was done, whereas summary tells you how many samples were taken at each plot ID.

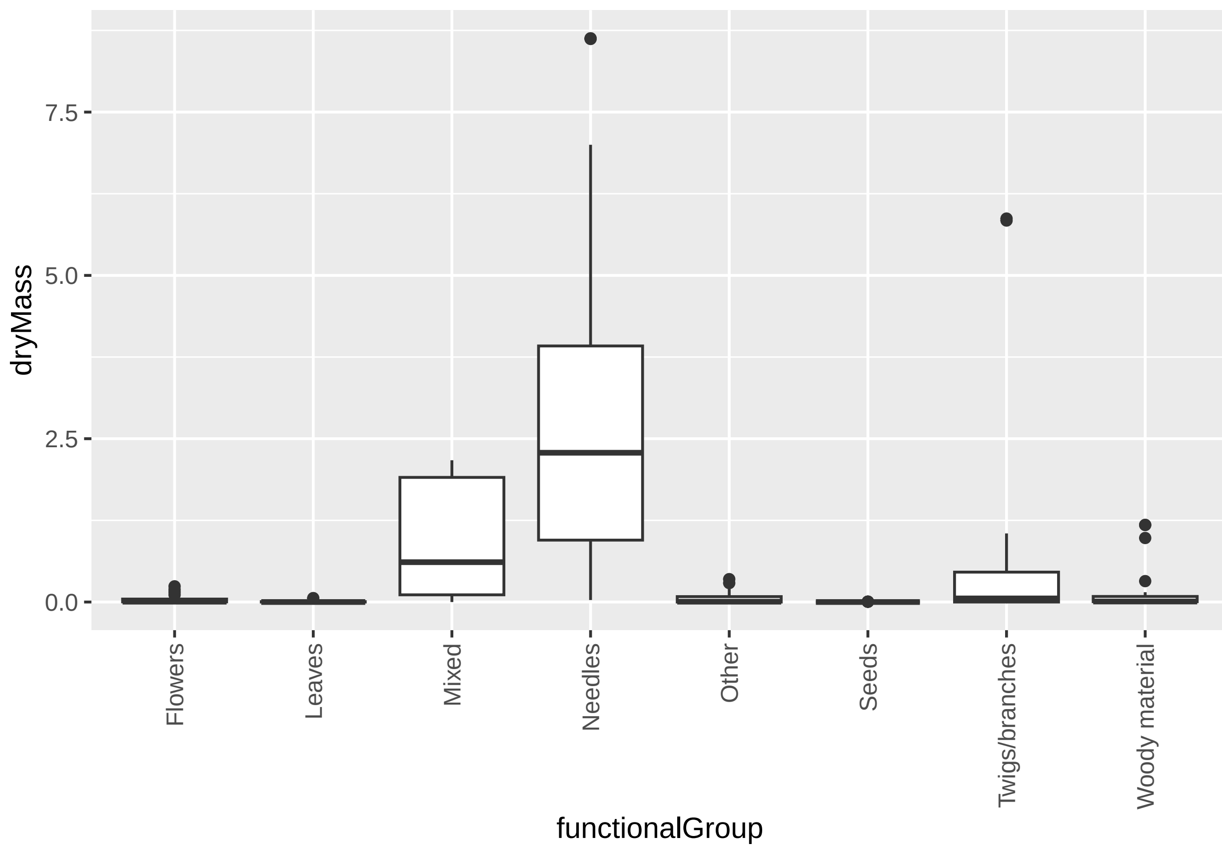
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```r
ggplot(data=Litter, aes(x=functionalGroup)) +
  geom_bar() + theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```

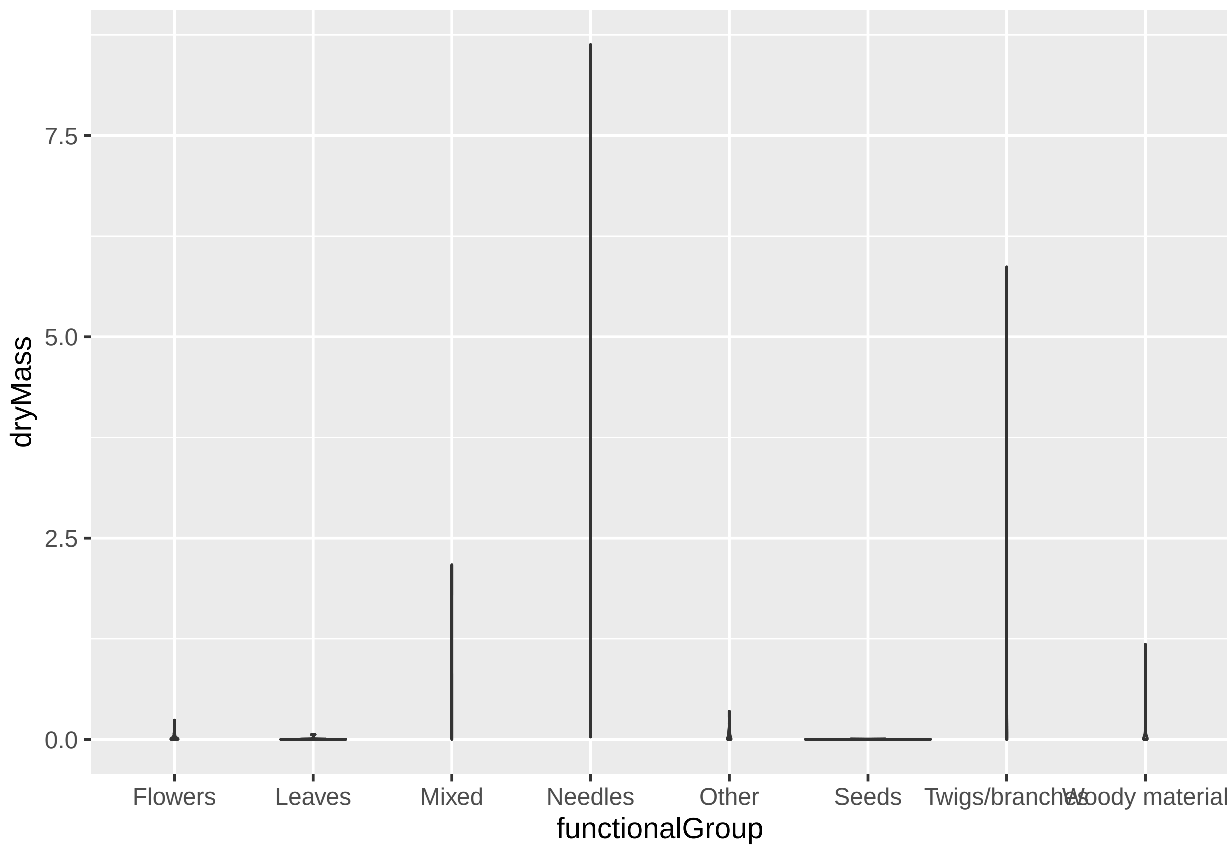


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of dryMass by functionalGroup.

```
ggplot(Litter) +
  geom_boxplot(aes(x=functionalGroup, y=dryMass)) + theme(axis.text.x = element_text(angle = 90, vjust =
```




```
ggplot(Litter) +
  geom_violin(aes(x=functionalGroup, y=dryMass))
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

> Answer:The boxplot also highlighted the large spread or variation among the data, particularly for Needles, mixed and slightly twigs/branches

What type(s) of litter tend to have the highest biomass at these sites?

> Answer:Needles